

Inference-Time Compute Hackathon (Applied AI Track)

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Context:

As frontier labs buy more [post-training data](#) from [domain experts](#) we need ways to **cheaply** understand if the data will improve downstream model performance.

Problem statement:

What **task-level** and **dataset-level** metrics correlate with a model's post-RL performance gain? ***Why?*** Where do these metrics sit on the cost-quality [Pareto frontier](#)?

(Hint: We are more interested in your rationale for selecting your metrics!)

In solving this problem you will likely:

1. Run experiments and post-train an open source LLM
2. Index the frontier of reinforcement learning literature
3. Create pipelines that generate high quality synthetic data
4. Contribute to the frontier of understanding data that is used to train current frontier models

Definitions/FAQs:

Q: What are some of the **primitives** that we can work with?

Usually we like to define **primitives** to understand the space of metrics we can define. For example, here are some primitives we find useful for deriving metrics:

T: string = task (e.g. "What is 2+2")

M: F(string) -> string = model (e.g. Opus 4.8 high)

V: F(string) -> {string: int} = verifier (e.g. run unit tests/grader/rubric on some string (usually model output) → per-criterion scores, e.g. {"tests_passed": 5, "compiles": 1})

y: string = a completion = M(T)

r: F({string:int}) -> float = reward/reduce

So one "scored rollout" could be: $r(V(M(T))) \rightarrow \text{float}$.

Note that these don't have to be your primitives and we encourage you to define ones of your own!

Q: What is a **task-level** metric?

A **task-level** metric is a measure which is produced by applying some function F to a set of primitives (of your choosing!).

Notice that **task-level** metrics can either be **model-agnostic** or **model-dependent**.

Examples of **task-level** on task $T = \text{"What is } 2+2\text{"}$:

1. Character length (e.g. $F_{\text{token_length}}(T, \text{primitives}[]) = 11$. Notice that this is **model-agnostic**!)
2. Model pass rate (e.g. $F_{\text{pass_rate}}(T, \text{primitives}[]) = 0.8$. Notice that this is **model-dependent**! [Example implementation of \$F\$ here](#))

Other (useful) task-level metrics we use: token length, reward variance, max score, branching factor, tool calls, reasoning steps.

*Tip: We find that **model-dependent** metrics are (usually) more helpful to us!*

Q: What is a **dataset-level** metric?

A **dataset-level** metric is a measure which is produced by applying some function F to a set of tasks $\{T_0, T_1, \dots, T_N\}$ and primitives.

Examples of **task-level** on dataset $D = \{T_0 = \text{"What is } 2+2\text{"}, T_1 = \text{"What is } 5 * 5\text{"}, T_2 = \text{"What is } 5-3\text{"}, T_3 = \text{"Why is the sky blue?"}\}$:

3. Task diversity (e.g. $F_{\text{task diversity}}(D, \text{primitives}[]) = 0.2$ (low coverage). This is an *important* metric. If interested I recommend checking out [this paper](#) and [this paper](#))
4. Average pass rate (e.g. $F_{\text{average pass rate}} = \text{mean}([F_{\text{pass_rate}}(T, \text{primitives}[]) \text{ for } T \text{ in } D]) = 0.2$. Notice that you can [reduce](#) over **task-level** metrics to achieve **dataset-level** metrics!)

Other dataset-level metrics you may want to consider:

[reducible holdout loss/learnability](#), [trajectory entropy](#), [intermediate-difficulty](#), [policy entropy](#), [Vendi diversity score](#), [redundancy](#), [learning-impact](#), [online difficulty filtering](#)

*Hint: Generally, many papers that deal with RL have some method they use to rank/select the data they train on. These methods *may prove useful* for defining both your **task-level** and **dataset-level** metrics!*

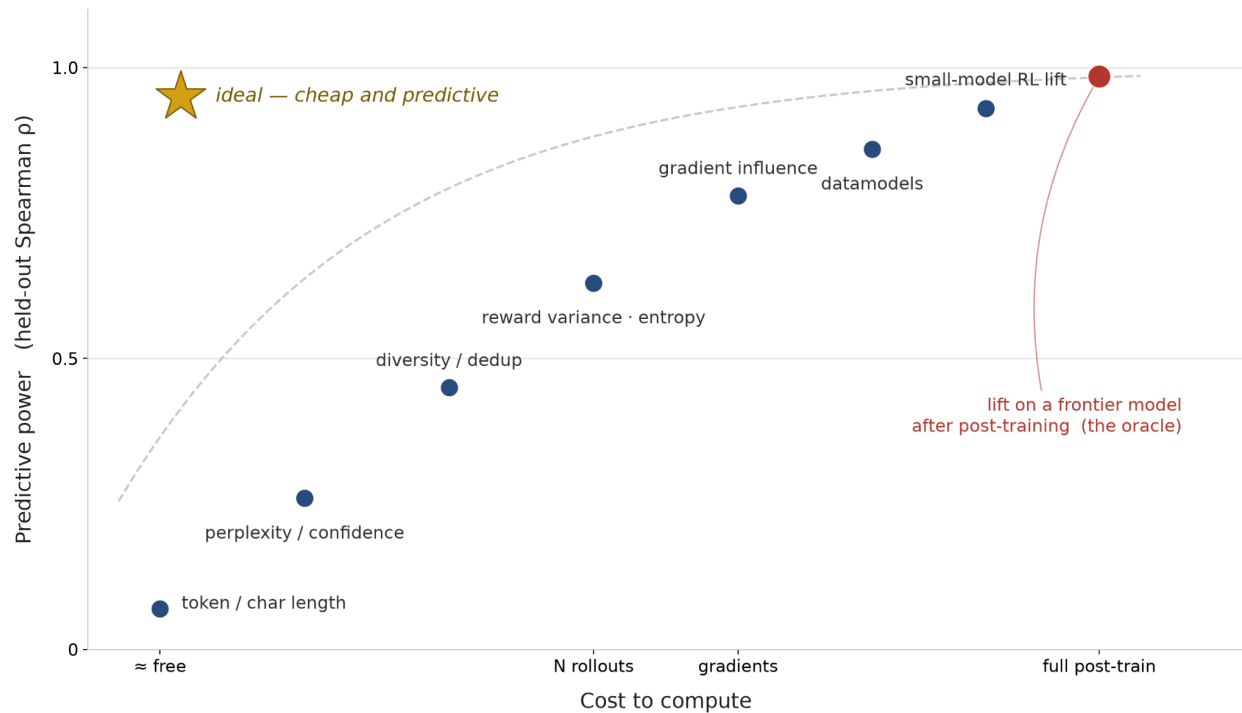
Q: What do you mean by cost-quality Pareto frontier?

We imagine that there exist a spectrum of metrics that range in both cost to generate and quality of predictive power. Intuitively you might expect that metrics that are *cheap* have little power to predict a model's post-RL performance gain and the opposite for expensive metrics!

However, for certain domains it may be the case that there are metrics that break the frontier!

To get a better understanding of what we mean by the cost-quality [Pareto frontier](#) we created this graph (which isn't necessarily accurate):

Cost vs. predictive power of data-quality metrics



Task Description:

While the following are the *recommended* guidelines for deriving your **task-level** and **dataset-level** metrics we will accept any solution that reasons from first principles!

These recommendations are based off of the following assumptions:

Resources: 8× H100 (80GB), single node · Anthropic API credits (\$100/participant) · Cognition API access (credit amount TBC) · Prime Intellect \$100 compute credits + open RL stack (verifiers, prime-rl, Environments Hub).

1. **Pick a model + benchmark** Any small open-source model + an easy, freely-available benchmark — math (GSM8K) or code (MBPP+/HumanEval+). Pick what's fast to train and verify.
2. **Partition the data into cohorts** Slice the task pool into several cohorts, holding size fixed and varying one property at a time. We recommend many cohorts that are both real and synthetic to ablate data quality!
3. **Post-train (RL) the model on each cohort** One GRPO run per cohort, same config across all of them (so cohort is the only variable).

4. **Measure lift** Eval each trained model on the benchmark; lift = accuracy(after) – accuracy(before).

5. **Define metrics on each cohort.** Propose per-task and dataset-level metrics — computed cheaply, with no training — that you think predict the lift with reasoning from first principles.

6. **Measure how well your metric predicts the lift on held-out cohorts.** Fit metric → lift, then quantify how far the predicted lift is from the actual lift on cohorts you didn't fit on. Any standard goodness-of-fit works (variance explained / R^2 , or a prediction error like RMSE, or rank correlation). Lower deviation = the metric genuinely tracks training value. Show the scatter; with few cohorts, holdout honesty matters more than the exact statistic.